

# Inference in Bayes Nets

- ▶ Given a Bayes net

$$\Pr(X_1, X_2, \dots, X_n) = \Pr(X_n \mid \text{Par}(X_n)) \Pr(X_{n-1} \mid \text{Par}(X_{n-1})) \cdots \Pr(X_1 \mid \text{Par}(X_1))$$

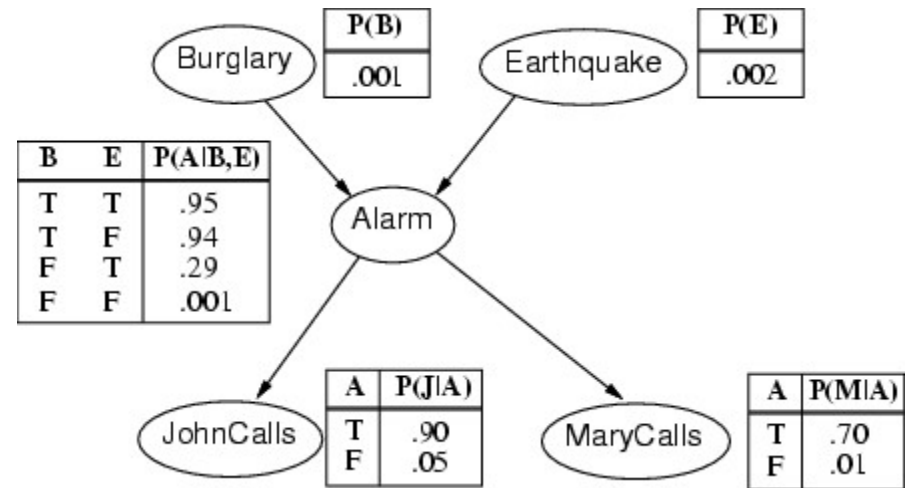
- ▶ And some **evidence**  $E = \{\text{specific known values for some of the variables}\}$  we want to compute the new probability distribution  $\Pr(X_k \mid E)$
- ▶ That is, we want to figure out  $\Pr(X_k = d \mid E)$  for all  $d \in \text{Dom}[X_k]$

# Inference in Bayes Nets

- ▶ E.g., computing probability of different diseases given symptoms, computing probability of hail storms given different metrological evidence, etc.
- ▶ In such cases getting a good estimate of the probability of the unknown event allows us to respond more effectively (gamble rationally)

# Inference in Bayes Nets

- ▶ In the Alarm example:



- ▶  $\Pr(\text{Burglary}, \text{Earthquake}, \text{Alarm}, \text{JohnCalls}, \text{MaryCalls}) =$   
 $\Pr(\text{Earthquake}) * \Pr(\text{Burglary}) * \Pr(\text{Alarm}|\text{Earthquake}, \text{Burglary}) * \Pr(\text{JohnCalls}|\text{Alarm}) * \Pr(\text{MaryCalls}|\text{Alarm})$
- ▶ And, e.g., we want to compute things like  $\Pr(\text{Burglary}=\text{True} | \text{MaryCalls}=\text{false}, \text{JohnCalls}=\text{true})$

# Variable Elimination

- ▶ Variable elimination uses the product decomposition that defines the Bayes Net and the summing out rule to compute posterior probabilities from the information (CPTs) already in the network.

## Example (Binary valued Variables)

$\Pr(A,B,C,D,E,F,G,H,I,J,K) =$

$\Pr(A)$

$\times \Pr(B)$

$\times \Pr(C|A)$

$\times \Pr(D|A,B)$

$\times \Pr(E|C)$

$\times \Pr(F|D)$

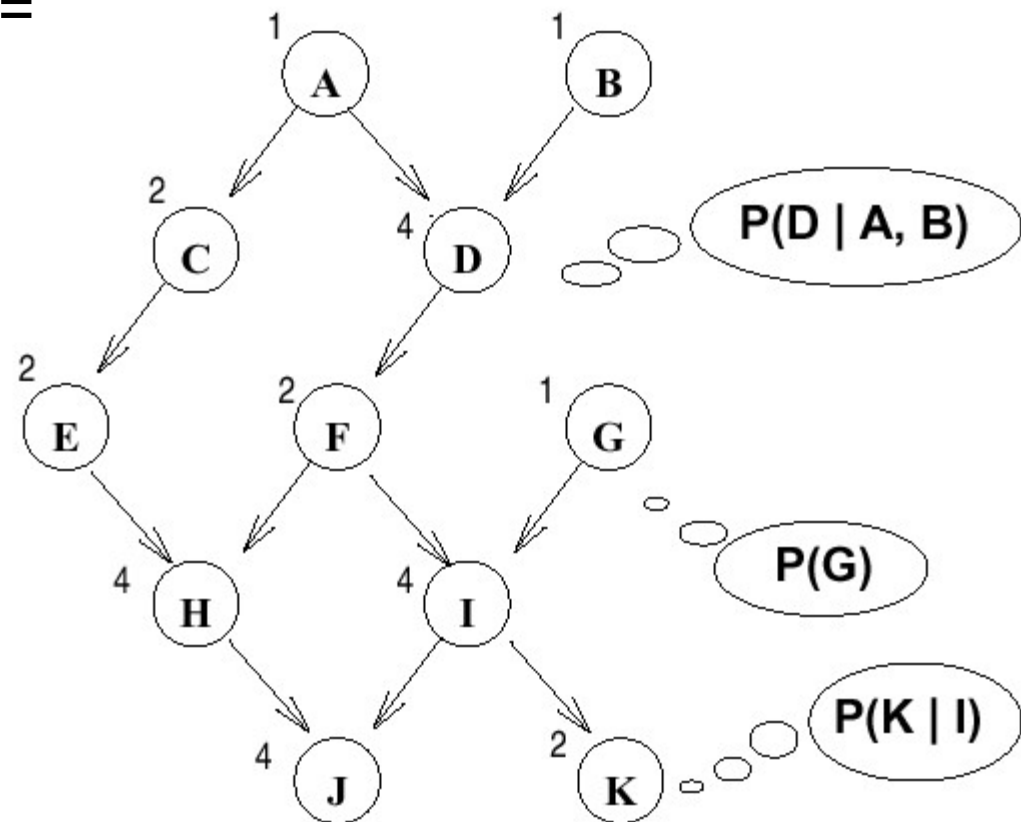
$\times \Pr(G)$

$\times \Pr(H|E,F)$

$\times \Pr(I|F,G)$

$\times \Pr(J|H,I)$

$\times \Pr(K|I)$



# Example

$$\begin{aligned} \Pr(A, B, C, D, E, F, G, H, I, J, K) = \\ \Pr(A) \Pr(B) \Pr(C | A) \Pr(D | A, B) \Pr(E | C) \Pr(F | D) \Pr(G) \\ \Pr(H | E, F) \Pr(I | F, G) \Pr(J | H, I) \Pr(K | I) \end{aligned}$$

Say that  $E = \{H=\text{true}, I=\text{false}\}$ , and we want to know

$$\Pr(D | h, -i) \quad (h: H \text{ is true, } -h: H \text{ is false})$$

$$\Pr(d | h, -i) = \Pr(d, h, -i) / \Pr(h, -i)$$

$$\Pr(-d | h, -i) = \Pr(-d, h, -i) / \Pr(h, -i)$$

So we are computing  $\Pr(d, h, -i)$  and  $\Pr(-d, h, -i)$  divided by their sum:

$$\Pr(d, h, -i) / N \quad \text{and} \quad \Pr(-d, h, -i) / N$$

$$N = \Pr(d, h, -i) + \Pr(-d, h, -i) = \Pr(h, -i)$$

This the same as **normalizing** the vector  $[\Pr(d, h, -i), \Pr(-d, h, -i)]$ .

# Example

First, we write as a sum for each value of D (i.e.  $D = d$  and  $D = -d$ )

- ▶  $\sum_{A,B,C,E,F,G,J,K} \Pr(A, B, C, d, E, F, h, -i, J, K) = \Pr(d, h, -i)$
- ▶  $\sum_{A,B,C,E,F,G,J,K} \Pr(A, B, C, -d, E, F, h, -i, J, K) = \Pr(-d, h, -i)$

Start by computing  $P(d,h,-i)$ . Use Bayes Net product decomposition to rewrite the summation:

- ▶  $\sum_{A,B,C,E,F,G,J,K} \Pr(A, B, C, d, E, F, h, -i, J, K) =$   
 $\sum_{A,B,C,E,F,G,J,K} \Pr(A) \Pr(B) \Pr(C | A) \Pr(d | A, B) \Pr(E | C)$   
 $\Pr(F | d) \Pr(G) \Pr(h | E, F) \Pr(-i | F, G) \Pr(J | h, -i)$   
 $\Pr(K | -i)$

## Example

Next, rearrange summations so that we are not summing over variables that do not depend on the summed variable.

$$= \sum_A \sum_B \sum_C \sum_E \sum_F \sum_G \sum_J \sum_K \Pr(A) \Pr(B) \Pr(C | A) \Pr(d | A, B) \\ \Pr(E | C) \Pr(F | d) \Pr(G) \Pr(h | E, F) \Pr(-i | F, G) \Pr(J | h, -i) \\ \Pr(K | -i)$$

$$= \sum_A \Pr(A) \sum_B \Pr(B) \sum_C \Pr(C | A) \Pr(d | A, B) \sum_E \Pr(E | C) \\ \sum_F \Pr(F | d) \sum_G \Pr(G) \Pr(h | E, F) \Pr(-i | F, G) \sum_J \Pr(J | h, -i) \\ \sum_K \Pr(K | -i)$$

$$= \sum_A \Pr(A) \sum_B \Pr(B) \Pr(d | A, B) \sum_C \Pr(C | A) \sum_E \Pr(E | C) \\ \sum_F \Pr(F | d) \Pr(h | E, F) \sum_G \Pr(G) \Pr(-i | F, G) \sum_J \Pr(J | h, -i) \\ \sum_K \Pr(K | -i)$$



## Example

Now start computing from the last summation to the first.

$$\begin{aligned} & \sum_A \Pr(A) \sum_B \Pr(B) \Pr(d \mid A, B) \sum_C \Pr(C \mid A) \sum_E \Pr(E \mid C) \\ & \quad \sum_F \Pr(F \mid d) \Pr(h \mid E, F) \sum_G \Pr(G) \Pr(-i \mid F, G) \\ & \quad \sum_J \Pr(J \mid h, -i) \\ & \quad \sum_K \Pr(K \mid -i) \end{aligned}$$

$$\sum_K \Pr(K \mid -i) = \Pr(k \mid -i) + \Pr(-k \mid -i) = c_1$$

$$\begin{aligned} & \sum_A \Pr(A) \sum_B \Pr(B) \Pr(d \mid A, B) \sum_C \Pr(C \mid A) \sum_E \Pr(E \mid C) \\ & \quad \sum_F \Pr(F \mid d) \Pr(h \mid E, F) \sum_G \Pr(G) \Pr(-i \mid F, G) \\ & \quad \sum_J \Pr(J \mid h, -i) c_1 \end{aligned}$$

# Example

$$c_1 \sum_A \Pr(A) \sum_B \Pr(B) \Pr(d \mid A, B) \sum_C \Pr(C \mid A) \sum_E \Pr(E \mid C) \\ \sum_F \Pr(F \mid d) \Pr(h \mid E, F) \sum_G \Pr(G) \Pr(-i \mid F, G) \\ \sum_J \Pr(J \mid h, -i)$$

## ► Note:

1. We have a new expression that does not have the variable K. K has been **eliminated**.
2.  $c_1$  does not depend on any of the variables so we can move it to the front

## Example

$$c_1 \sum_A \Pr(A) \sum_B \Pr(B) \Pr(d \mid A, B) \sum_C \Pr(C \mid A) \sum_E \Pr(E \mid C) \\ \sum_F \Pr(F \mid d) \Pr(h \mid E, F) \sum_G \Pr(G) \Pr(-i \mid F, G) \\ \sum_J \Pr(J \mid h, -i)$$

$$\sum_J \Pr(J \mid h, -i) = \Pr(j \mid h, -i) + \Pr(-j \mid h, -i) = c_2$$

Now we have **eliminated J**, again the sum does not depend on any variable:

$$c_1 c_2 \sum_A \Pr(A) \sum_B \Pr(B) \Pr(d \mid A, B) \sum_C \Pr(C \mid A) \sum_E \Pr(E \mid C) \\ \sum_F \Pr(F \mid d) \Pr(h \mid E, F) \sum_G \Pr(G) \Pr(-i \mid F, G)$$

# Example

$$c_1 c_2 \sum_A \Pr(A) \sum_B \Pr(B) \Pr(d \mid A, B) \sum_C \Pr(C \mid A) \sum_E \Pr(E \mid C) \\ \sum_F \Pr(F \mid d) \Pr(h \mid E, F) \sum_G \Pr(G) \Pr(-i \mid F, G)$$

$$\sum_G \Pr(G) \Pr(-i \mid F, G) = \Pr(g) \Pr(-i \mid F, g) + \Pr(-g) \Pr(-i \mid F, -g)$$

## ► Note:

1.  $\Pr(-i \mid F, g)$  depends on the value of  $F$ , so this is not a single number. So the sum depends on  $F$ .
2. Once  $F$  is fixed to  $f$  or  $-f$ , the sum yields a fixed number.
3. We introduce a function to represent this sum:

$$f_1(F) = \Pr(g) \Pr(-i \mid F, g) + \Pr(-g) \Pr(-i \mid F, -g)$$

$$f_1(f) = \Pr(g) \Pr(-i \mid f, g) + \Pr(-g) \Pr(-i \mid f, g)$$

$$f_1(-f) = \Pr(g) \Pr(-i \mid -f, g) + \Pr(-g) \Pr(-i \mid -f, g)$$

are fixed numbers

## Example

$$c_1 c_2 \sum_A \Pr(A) \sum_B \Pr(B) \Pr(d \mid A, B) \sum_C \Pr(C \mid A) \sum_E \Pr(E \mid C) \\ \sum_F \Pr(F \mid d) \Pr(h \mid E, F) f_1(F)$$

$$\sum_F \Pr(F \mid d) \Pr(h \mid E, F) f_1(F) = \\ \Pr(f \mid d) \Pr(h \mid E, f) f_1(f) + \Pr(-f \mid d) \Pr(h \mid E, -f) f_1(-f)$$

This sum depends on  $E$ . So we can once again introduce a new function to represent this sum. This new function depends on  $E$

$$f_2(E) = \Pr(f \mid d) \Pr(h \mid E, f) f_1(f) + \Pr(-f \mid d) \Pr(h \mid E, -f) f_1(-f)$$

$f_2(e)$  and  $f_2(-e)$  are fixed numbers

## Example

- ▶ We can continue this way eliminating one variable after another. Until we sum out  $A$  to obtain a single number equal to  $\Pr(d, h, -i)$
- ▶ A similar computation produces  $\Pr(-d, h, -i)$
- ▶ Normalizing these two numbers gives us  $\Pr(d \mid h, i)$  and  $\Pr(-d \mid h, i)$
- ▶ Alternatively, we can keep  $D$  as a variable, and compute a function of  $D$ ,  $f_k(D)$  whose two values  $f_k(d)$  and  $f_k(-d)$  are the values we want.

# Variable Elimination (VE)

- ▶ This process is called **variable elimination**.
- ▶ By computing the intermediate functions  $f_1(F)$ ,  $f_2(E)$  etc. we are actually **storing values** that we can reuse many times during the computation.
- ▶ In this way variable elimination is a form of *dynamic programming*, where we save sub-computations to avoid re-computations.

# Variable Elimination (VE)

- ▶ In general, at each stage VE will **sum out** the innermost variable, computing a new **function** over the **variables in that sum**.
- ▶ The function specifies one number for each different **instantiation** of its arguments.
- ▶ We store these functions as a **table** with one entry per instantiation of the variables
- ▶ The size of these tables is **exponential** in the *number of variables appearing in the sum*, e.g.,

$$\sum_F \Pr(F|D) \Pr(h|E,F)t(F)$$

depends on the value of D and E, thus we will obtain  $|\text{Dom}[D]| * |\text{Dom}[E]|$  different numbers in the resulting table.



# Factors

- ▶ We call these **tables of values** computed by VE **factors**.
- ▶ Note that the original probabilities that appear in the summation, e.g.,  $P(C|A)$ , are also tables of values (one value for each instantiation of  $C$  and  $A$ ).  
Thus we also call the **original CPTs factors**.
- ▶ Each factor is a function of some variables
  - ▶ E.g., if  $f(A,C)$  is associated with  $P(C|A)$ ,  $f(A,C)$  maps each value of its arguments to a probability number.
  - ▶ A tabular representation is exponential in the number of arguments of the factor.

# Operations on Factors

- ▶ If we examine the inside-out summation process we see that various operations occur on factors.
- ▶ We can specify the algorithm for Variable Elimination by precisely specifying these operations.
- ▶ Notation:  $f(\mathbf{X}, \mathbf{Y})$  denotes a factor over the variables  $\mathbf{X} \cup \mathbf{Y}$  where  $\mathbf{X}$  and  $\mathbf{Y}$  are **sets** of variables.

# The Product of Two Factors

- ▶ Let  $f(\mathbf{X}, \mathbf{Y})$  &  $g(\mathbf{Y}, \mathbf{Z})$  be two factors with variables  $\mathbf{Y}$  in common
- ▶ The *product* of  $f$  and  $g$ , denoted  $h = f * g$  (or sometimes just  $h = fg$ ), is defined:

$$h(\mathbf{X}, \mathbf{Y}, \mathbf{Z}) = f(\mathbf{X}, \mathbf{Y}) \times g(\mathbf{Y}, \mathbf{Z})$$

| f(A,B) |     | g(B,C) |     | h(A,B,C) |      |        |      |
|--------|-----|--------|-----|----------|------|--------|------|
| ab     | 0.9 | bc     | 0.7 | abc      | 0.63 | ab~c   | 0.27 |
| a~b    | 0.1 | b~c    | 0.3 | a~bc     | 0.08 | a~b~c  | 0.02 |
| ~ab    | 0.4 | ~bc    | 0.8 | ~abc     | 0.28 | ~ab~c  | 0.12 |
| ~a~b   | 0.6 | ~b~c   | 0.2 | ~a~bc    | 0.48 | ~a~b~c | 0.12 |

# Summing a Variable Out of a Factor

- ▶ Let  $f(X, \mathbf{Y})$  be a factor with variable  $X$  ( $\mathbf{Y}$  is a set)
- ▶ We *sum out* variable  $X$  from  $f$  to produce a new factor  $h = \sum_X f$ , which is defined:

$$h(\mathbf{Y}) = \sum_{x \in \text{Dom}(X)} f(x, \mathbf{Y})$$

| f(A,B) |     | h(B) |     |
|--------|-----|------|-----|
| ab     | 0.9 | b    | 1.3 |
| a~b    | 0.1 | ~b   | 0.7 |
| ~ab    | 0.4 |      |     |
| ~a~b   | 0.6 |      |     |

# Restricting a Factor

- ▶ Let  $f(X, Y)$  be a factor with variable  $X$  ( $Y$  is a set)
- ▶ We *restrict* factor  $f$  *to*  $X=a$  by setting  $X$  to the value  $x$  and “deleting” incompatible elements of  $f$ ’s domain .  
Define  $h = f_{X=a}$  as:  $h(Y) = f(a, Y)$

| $f(A, B)$      |     | $h(B) = f_{A=a}$ |     |
|----------------|-----|------------------|-----|
| $ab$           | 0.9 | $b$              | 0.9 |
| $a\sim b$      | 0.1 | $\sim b$         | 0.1 |
| $\sim ab$      | 0.4 |                  |     |
| $\sim a\sim b$ | 0.6 |                  |     |

# Variable Elimination the Algorithm

Given query var  $Q$ , evidence vars  $\underline{E}$  (set of variables observed to have values  $\underline{e}$ ), remaining vars  $\mathbf{Z}$ . Let  $F$  be the set of original CPTs.

1. Replace each factor  $f \in F$  that mentions a variable(s) in  $\underline{E}$  with its **restriction**  $f_{\underline{E}=\underline{e}}$  (this might yield a factor over no variables, a constant)
2. For each  $Z_j$ —in the order given—eliminate  $Z_j \in \mathbf{Z}$  as follows:
  - (a) Compute **new factor**  $g_j = \sum_{Z_j} f_1 * f_2 * \dots * f_k$ ,  
where the  $f_i$  are the factors in  $F$  that include  $Z_j$
  - (b) **Remove** the factors  $f_i$  that mention  $Z_j$  from  $F$  and  
add new factor  $g_j$  to  $F$
3. The remaining factors refer only to the query variable  $Q$ .  
**Take their product and normalize** to produce  $\Pr(Q|\underline{e})$

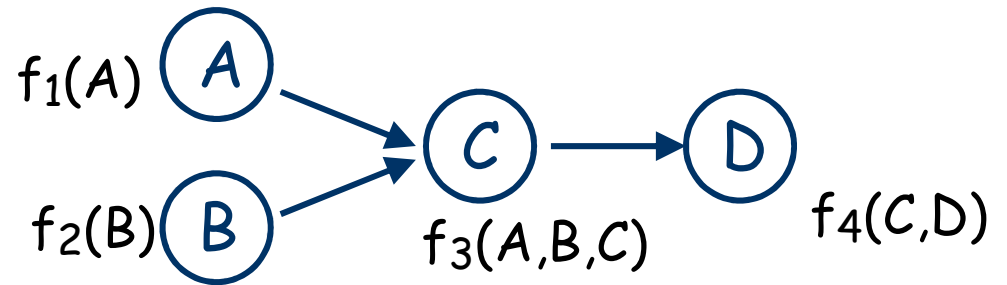
# VE: Example

**Factors:**  $f_1(A)$   $f_2(B)$   $f_3(A,B,C)$   
 $f_4(C,D)$

**Query:** A

*Evidence:*  $D = d$

**Elim. Order:** C, B



1. Restriction: replace  $f_4(C,D)$  with  $f_5(C) = f_4(C,d)$

2.1. Compute & Add  $f_6(A,B) = \sum_C f_5(C) f_3(A,B,C)$

Remove:  $f_3(A,B,C)$ ,  $f_5(C)$

2.2. Compute & Add  $f_7(A) = \sum_B f_6(A,B) f_2(B)$

Remove:  $f_6(A,B)$ ,  $f_2(B)$

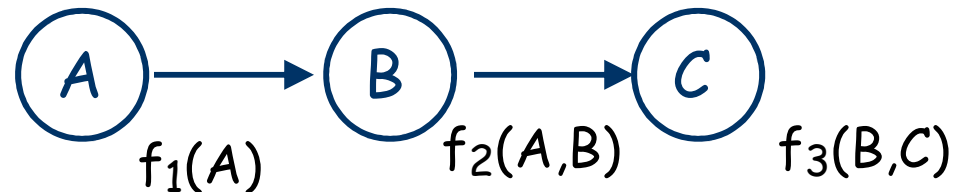
Last factors:  $f_7(A)$ ,  $f_1(A)$ . The product  $f_1(A) \times f_7(A)$  is un-normalized. So:

$P(A|d) = \alpha f_1(A) \times f_7(A)$

where  $\alpha = 1/\sum_A f_1(A)f_7(A)$

# Numeric Example

- Query: C
- No Evidence



| $f_1(A)$ |     | $f_2(A,B)$ |     | $f_3(B,C)$ |     | $f_4(B)$<br>$\sum_A f_2(A,B) f_1(A)$ |      | $f_5(C)$<br>$\sum_B f_3(B,C) f_4(B)$ |       |
|----------|-----|------------|-----|------------|-----|--------------------------------------|------|--------------------------------------|-------|
| a        | 0.9 | ab         | 0.9 | bc         | 0.7 | b                                    | 0.85 | c                                    | 0.625 |
| ~a       | 0.1 | a~b        | 0.1 | b~c        | 0.3 | ~b                                   | 0.15 | ~c                                   | 0.375 |
|          |     | ~ab        | 0.4 | ~bc        | 0.2 |                                      |      |                                      |       |
|          |     | ~a~b       | 0.6 | ~b~c       | 0.8 |                                      |      |                                      |       |



# Numeric Example

| $f_1(A)$ |     | $f_2(A,B)$     |     | $f_3(B,C)$     |     | $f_4(B)$<br>$\sum_A f_2(A,B)f_1(A)$ |      | $f_5(C)$<br>$\sum_B f_3(B,C) f_4(B)$ |       |
|----------|-----|----------------|-----|----------------|-----|-------------------------------------|------|--------------------------------------|-------|
| a        | 0.9 | ab             | 0.9 | bc             | 0.7 | b                                   | 0.85 | c                                    | 0.625 |
| $\sim a$ | 0.1 | $a\sim b$      | 0.1 | $b\sim c$      | 0.3 | $\sim b$                            | 0.15 | $\sim c$                             | 0.375 |
|          |     | $\sim ab$      | 0.4 | $\sim bc$      | 0.2 |                                     |      |                                      |       |
|          |     | $\sim a\sim b$ | 0.6 | $\sim b\sim c$ | 0.8 |                                     |      |                                      |       |

$$\begin{aligned}
 f_4(b) &= \sum_A f_2(A,b)f_1(A) \\
 &= f_2(a,b)f_1(a) + f_2(\sim a,b)f_1(\sim a) \\
 &= 0.9*0.9 + 0.4 * 0.1 = 0.85
 \end{aligned}$$

# Numeric Example

| $f_1(A)$ |     | $f_2(A,B)$     |     | $f_3(B,C)$     |     | $f_4(B)$<br>$\sum_A f_2(A,B)f_1(A)$ |      | $f_5(C)$<br>$\sum_B f_3(B,C) f_4(B)$ |       |
|----------|-----|----------------|-----|----------------|-----|-------------------------------------|------|--------------------------------------|-------|
| a        | 0.9 | ab             | 0.9 | bc             | 0.7 | b                                   | 0.85 | c                                    | 0.625 |
| $\sim a$ | 0.1 | $a\sim b$      | 0.1 | $b\sim c$      | 0.3 | $\sim b$                            | 0.15 | $\sim c$                             | 0.375 |
|          |     | $\sim ab$      | 0.4 | $\sim bc$      | 0.2 |                                     |      |                                      |       |
|          |     | $\sim a\sim b$ | 0.6 | $\sim b\sim c$ | 0.8 |                                     |      |                                      |       |

$$\begin{aligned}
 f_4(\sim b) &= \sum_A f_2(A, \sim b) f_1(A) \\
 &= f_2(a, \sim b) f_1(a) + f_2(\sim a, \sim b) f_1(\sim a) \\
 &= 0.1 * 0.9 + 0.6 * 0.1 = 0.15
 \end{aligned}$$

Note:  $f_4(b) + f_4(\sim b) = 1$ . But the intermediate factors are not always probabilities.

# Numeric Example

| $f_1(A)$ |     | $f_2(A,B)$     |     | $f_3(B,C)$     |     | $f_4(B)$<br>$\sum_A f_2(A,B)f_1(A)$ |      | $f_5(C)$<br>$\sum_B f_3(B,C) f_4(B)$ |       |
|----------|-----|----------------|-----|----------------|-----|-------------------------------------|------|--------------------------------------|-------|
| a        | 0.9 | ab             | 0.9 | bc             | 0.7 | b                                   | 0.85 | c                                    | 0.625 |
| $\sim a$ | 0.1 | $a\sim b$      | 0.1 | $b\sim c$      | 0.3 | $\sim b$                            | 0.15 | $\sim c$                             | 0.375 |
|          |     | $\sim ab$      | 0.4 | $\sim bc$      | 0.2 |                                     |      |                                      |       |
|          |     | $\sim a\sim b$ | 0.6 | $\sim b\sim c$ | 0.8 |                                     |      |                                      |       |

$$\begin{aligned}
 f_5(c) &= \sum_B f_3(B,c)f_4(B) \\
 &= f_3(b,c)f_4(b) + f_3(\sim b,c)f_4(\sim b) \\
 &= 0.7*0.85 + 0.2*0.15 = 0.625
 \end{aligned}$$

# Numeric Example

| $f_1(A)$ |     | $f_2(A,B)$     |     | $f_3(B,C)$     |     | $f_4(B)$<br>$\sum_A f_2(A,B)f_1(A)$ |      | $f_5(C)$<br>$\sum_B f_3(B,C) f_4(B)$ |       |
|----------|-----|----------------|-----|----------------|-----|-------------------------------------|------|--------------------------------------|-------|
| a        | 0.9 | ab             | 0.9 | bc             | 0.7 | b                                   | 0.85 | c                                    | 0.625 |
| $\sim a$ | 0.1 | $a\sim b$      | 0.1 | $b\sim c$      | 0.3 | $\sim b$                            | 0.15 | $\sim c$                             | 0.375 |
|          |     | $\sim ab$      | 0.4 | $\sim bc$      | 0.2 |                                     |      |                                      |       |
|          |     | $\sim a\sim b$ | 0.6 | $\sim b\sim c$ | 0.8 |                                     |      |                                      |       |

$$\begin{aligned}
 f_5(\sim c) &= \sum_B f_3(B, \sim c) f_4(B) \\
 &= f_3(b, \sim c) f_4(b) + f_3(\sim b, \sim c) f_4(\sim b) \\
 &= 0.3 * 0.85 + 0.8 * 0.15 = 0.375
 \end{aligned}$$

# Numeric Example

| $f_1(A)$ |     | $f_2(A,B)$     |     | $f_3(B,C)$     |     | $f_4(B)$<br>$\sum_A f_2(A,B)f_1(A)$ |      | $f_5(C)$<br>$\sum_B f_3(B,C) f_4(B)$ |       |
|----------|-----|----------------|-----|----------------|-----|-------------------------------------|------|--------------------------------------|-------|
| a        | 0.9 | ab             | 0.9 | bc             | 0.7 | b                                   | 0.85 | c                                    | 0.625 |
| $\sim a$ | 0.1 | $a\sim b$      | 0.1 | $b\sim c$      | 0.3 | $\sim b$                            | 0.15 | $\sim c$                             | 0.375 |
|          |     | $\sim ab$      | 0.4 | $\sim bc$      | 0.2 |                                     |      |                                      |       |
|          |     | $\sim a\sim b$ | 0.6 | $\sim b\sim c$ | 0.8 |                                     |      |                                      |       |

$f_5(C)$  is already normalized

$$\Pr(c) = 0.625$$

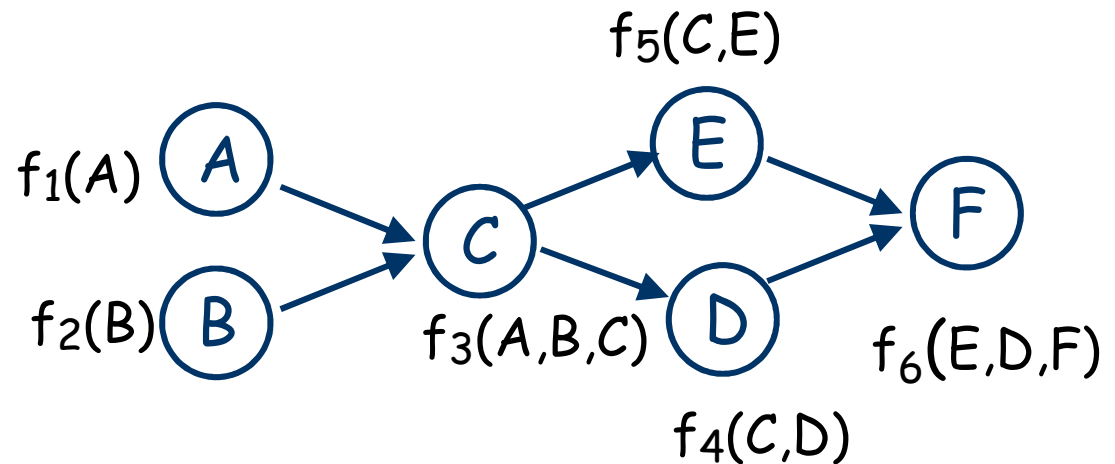
$$\Pr(\sim c) = 0.375$$

# VE: Buckets as a Notational Device

Query: D

No Evidence

Ordering:  
C, F, A, B, E, D



1. C:

2. F:

3. A:

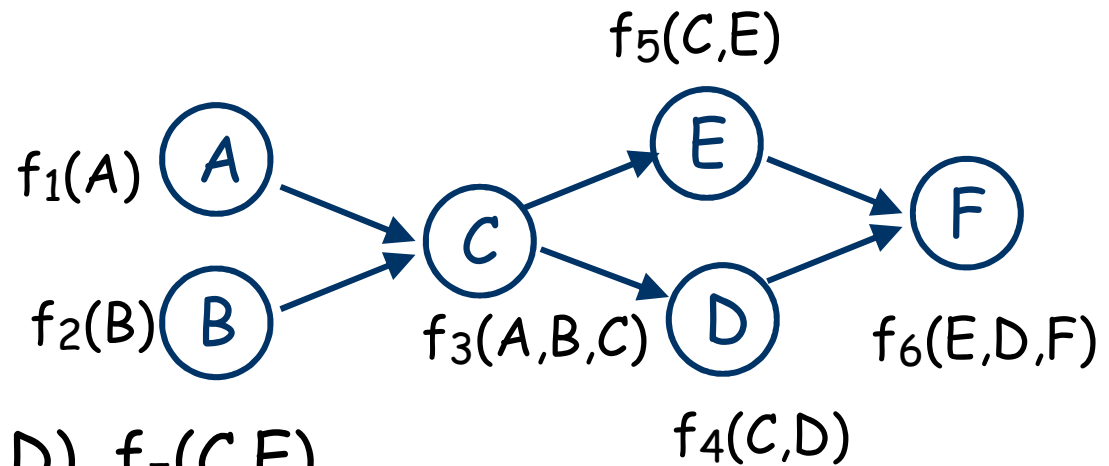
4. B:

5. E:

6. D:

## VE: Buckets—Place Original Factors in first bucket that contains one of its variables

Ordering:  
C, F, A, B, E, D



1. C:  $f_3(A, B, C)$ ,  $f_4(C, D)$ ,  $f_5(C, E)$

2. F:  $f_6(E, D, F)$

3. A:  $f_1(A)$

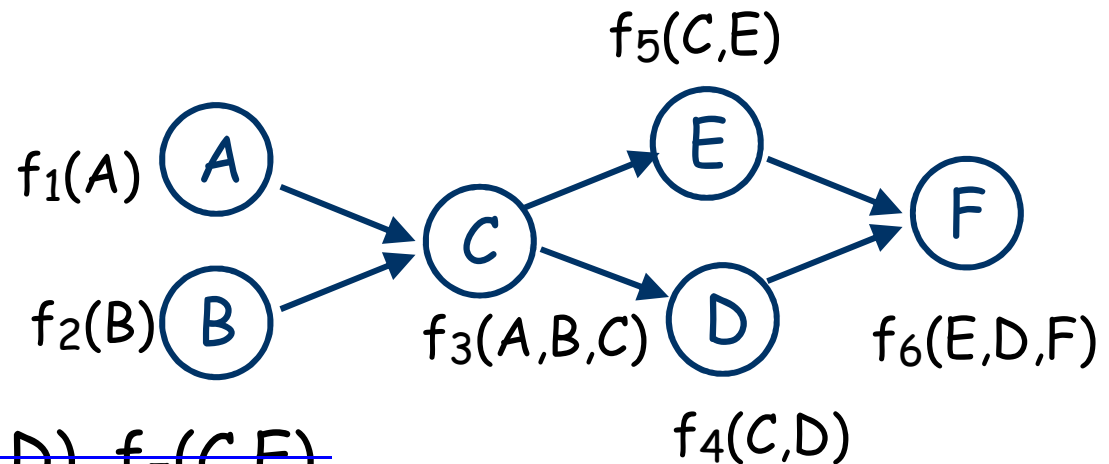
4. B:  $f_2(B)$

5. E:

6. D:

# VE: Buckets—Place Original Factors in first bucket that contains one of its variables

Ordering:  
C, F, A, B, E, D



1. ~~C:  $f_3(A,B,C)$ ,  $f_4(C,D)$ ,  $f_5(C,E)$~~

2. F:  $f_6(E,D,F)$

3. A:  $f_1(A)$ ,  $f_7(A,B,D,E)$

4. B:  $f_2(B)$

5. E:

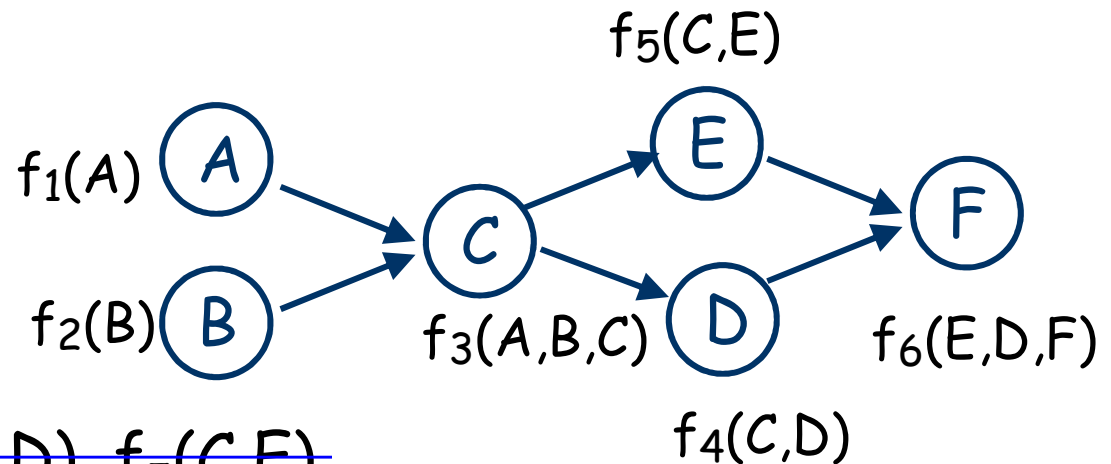
6. D:

$$1. \sum_C f_3(A,B,C) f_4(C,D) f_5(C,E) = f_7(A,B,D,E)$$



## VE: Buckets—Place Original Factors in first bucket that contains one of its variables

Ordering:  
C, F, A, B, E, D



1. ~~C:  $f_3(A,B,C)$ ,  $f_4(C,D)$ ,  $f_5(C,E)$~~

2. ~~F:  $f_6(E,D,F)$~~

$$2. \sum_F f_6(E,D,F) = f_8(E,D)$$

3. A:  $f_1(A)$ ,  $f_7(A,B,D,E)$

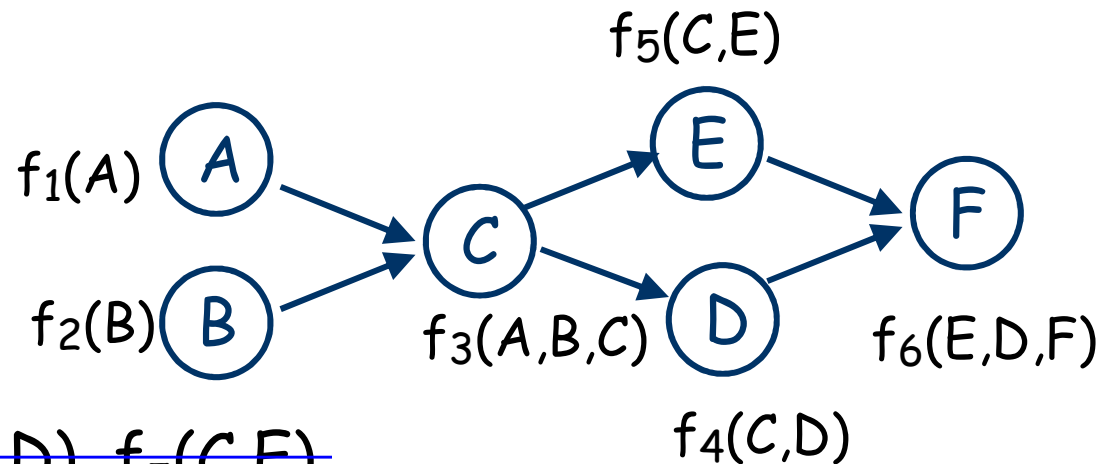
4. B:  $f_2(B)$

5. E:  $f_8(E,D)$

6. D:

## VE: Buckets—Place Original Factors in first bucket that contains one of its variables

Ordering:  
C, F, A, B, E, D



1. ~~C:  $f_3(A, B, C)$ ,  $f_4(C, D)$ ,  $f_5(C, E)$~~

2. ~~F:  $f_6(E, D, F)$~~

3. ~~A:  $f_1(A)$ ,  $f_7(A, B, D, E)$~~

$$3. \sum_A f_1(A) f_7(A, B, D, E) = f_9(B, D, E)$$

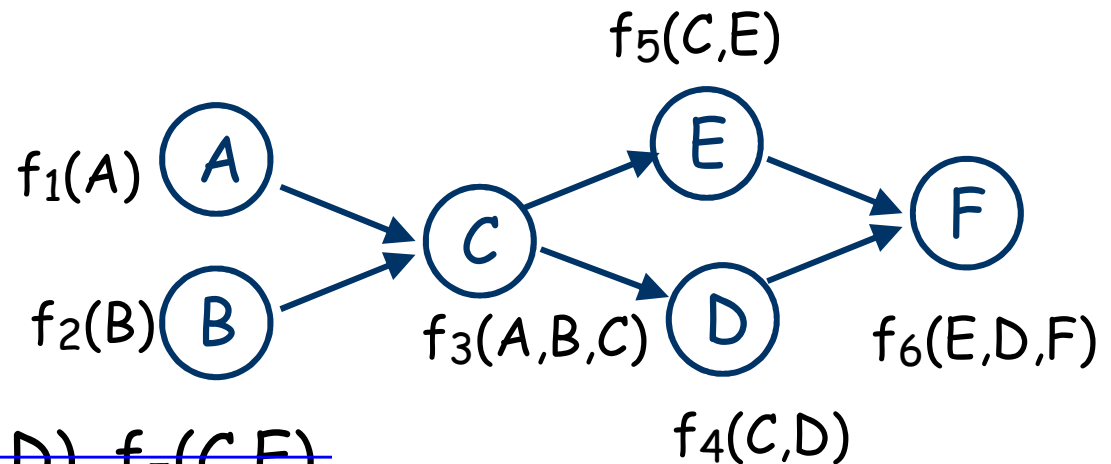
4. B:  $f_2(B)$ ,  $f_9(B, D, E)$

5. E:  $f_8(E, D)$

6. D:

## VE: Buckets—Place Original Factors in first bucket that contains one of its variables

Ordering:  
C, F, A, B, E, D



1. ~~C:  $f_3(A, B, C)$ ,  $f_4(C, D)$ ,  $f_5(C, E)$~~

2. ~~F:  $f_6(E, D, F)$~~

3. ~~A:  $f_1(A)$ ,  $f_7(A, B, D, E)$~~

4. ~~B:  $f_2(B)$ ,  $f_9(B, D, E)$~~

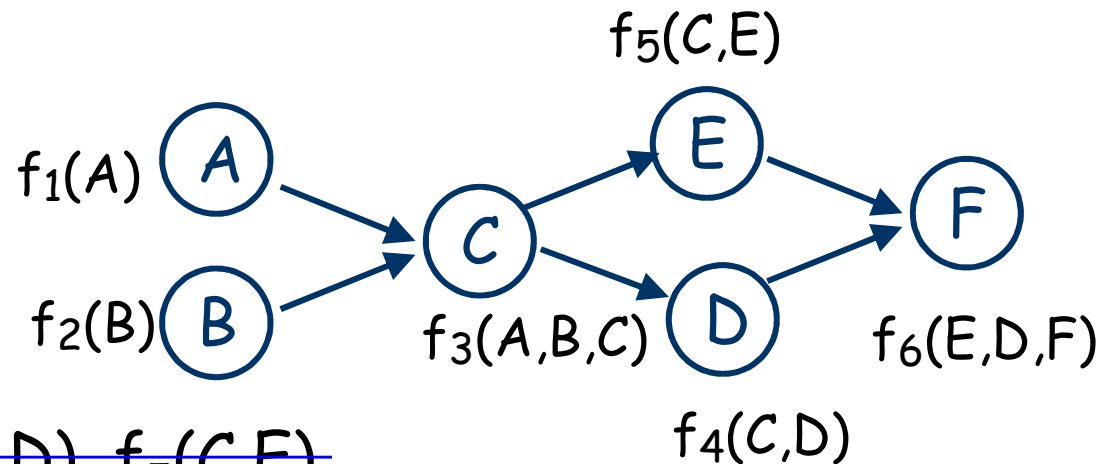
5. E:  $f_8(E, D)$ ,  $f_{10}(D, E)$

6. D:

$$4. \sum_B f_2(B) f_9(B, D, E) = f_{10}(D, E)$$

## VE: Buckets—Place Original Factors in first bucket that contains one of its variables

Ordering:  
C, F, A, B, E, D



1. ~~C:  $f_3(A, B, C)$ ,  $f_4(C, D)$ ,  $f_5(C, E)$~~

2. ~~F:  $f_6(E, D, F)$~~

3. ~~A:  $f_1(A)$ ,  $f_7(A, B, D, E)$~~

4. ~~B:  $f_2(B)$ ,  $f_9(B, D, E)$~~

5. ~~E:  $f_8(E, D)$ ,  $f_{10}(D, E)$~~

6. D:  $f_{11}(D)$

$$5. \sum_E f_8(E, D) f_{10}(D, E) = f_{11}(D)$$

$f_{11}$  is the final answer, once we normalize it.